Coexistence of evolutionary strategies

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Abstract

Introduction

Evolutionary game theory (EGT) provides a powerful framework for predicting which competing strategies are evolutionarily stable. Because the success of each strategy depends on the frequency of other strategies, even simple two-strategy games can yield heterogeneous outcomes, including the dominance of one strategy (prisoner's dilemma and harmony games), coexistence (hawk–dove games), and bistability (stag-hunt games). These differences are typically understood through payoff inequalities, invasion criteria, or stability analysis, but Tarnita and Traulsen (2025) recently argued that an ecological perspective on EGT may yield new, ecological insights. In particular, recasting game dynamics as a coexistence problem immediately raises two important questions: when two strategies coexist, what mechanism stabilizes it? And when coexistence fails, is it because stabilizing effects are absent, or because fitness differences are too large?

Modern coexistence theory (MCT) from community ecology presents a particularly useful framework for answering these questions. MCT predicts that coexistence of two competing entities, be they species or strategies, requires stabilizing niche differences to be greater than average fitness differences. In community ecology, stabilizing niche differences represent differences in factors that limit the growth of competing species, causing each species to limit itself more than it limits its competitor. In EGT, a strategy's niche can be interpreted as strategic environments generated by the composition of competing strategies and their interactions. Thus, stabilizing niche differences arise when strategies are favored by different strategic environments, such that the spread of each strategy creates conditions that limit its own relative growth and favor the recovery of its competitor. In contrast, fitness differences represent differences in the ability of each strategy to compete under the limiting strategic environments created by its competitor. This differs from payoff differences in EGT, which depend on the underlying strategy frequency and therefore combine both stabilizing effects caused by changes in the strategic environment and residual fitness differences that favor one strategy over another.

So far, MCT has been primarily utilized for understanding species coexistence, but recent work showed that MCT can be extended beyond species competition, to other competitive systems, allowing one to directly quantify niche and fitness differences of competing entities. Inspired by this work, we present a new theoretical framework for analyzing the coexistene and exclusion of competing strategies by quantifying their niche and fitness differences.

Mathematical theory

We begin with a classic replicator equation, which allows us to model changes in the frequency of competing strategies based on their payoffs:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \tag{1}$$

where x_i represents the frequency of strategy i, $\pi_i(\mathbf{x})$ represents the payoff of strategy i for a given strategy distribution $\mathbf{x}$, and $\bar{\pi}(\mathbf{x}) = \sum_i \pi_i(\mathbf{x})x_i$ represents the average payoff. Since strategy frequencies must sum to 1 (i.e., $\sum_i x_i = 1$), a single-strategy equilibrium dominated by strategy i can be described by a unit vector $\mathbf{e}_i$, where all elements are equal to zero except for the i-th element. We use these single-strategy equilibria to define niche and fitness differences between two competing strategies i and j. However, because payoffs in many games can be negative, payoff ratios are difficult to interpret. Thus, we work on a positive multiplicative scale by exponentiating the payoff terms:

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x})). \tag{2}$$

For convenience, we refer to $W_i(\mathbf{x})$ as the multiplicative payoff of strategy i.

First, we define the niche overlap ρ between strategies i and j as the ratio between the geometric mean of each strategy's multiplicative payoff in the strategic environment that it creates when common, $\sqrt{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}$, and the geometric mean of each strategy's multiplicative payoff in the environment generated by its competitor, $\sqrt{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}$. The corresponding niche difference is then given by

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \tag{3}$$

When $1 - \rho > 0$, each strategy performs better, on average, in the environment generated by its competitor than in the environment generated by itself. In other words, intraspecific competition is stronger than interspecific competition.

Second, we define the fitness difference as the ratio between the geometric means of the multiplicative payoff of each strategy across the two single-strategy environments:

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \tag{4}$$

In other words, strategy j has an average competitive advantage over strategy i ($\kappa_j/\kappa_i > 1$) when it performs better, on average, across the environments generated by itself and its competitor. Then, analogous to MCT, mutual invasion requires the fitness ratio to fall within the bounds set by niche overlap:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \tag{5}$$

This decomposition allows us to understand how different mechanisms contribute to the coexistence or exclusion of competing strategies.

Two strategy games

As an illustrative example, we begin with classic two-strategy games, which can be described by the following payoff matrix (Figure 1A):

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \tag{6}$$

Here, the first and second rows represent strategies 1 and 2, respectively, and the columns represent the strategy of the interacting opponent. Writing x_1 to denote the frequency of strategy 1, the expected payoffs are

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \tag{7}$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \tag{8}$$

Thus, the dynamics can be written as

$$\frac{dx_1}{dt} = x_1(1 - x_1)\left[(R - T)x_1 + (S - P)(1 - x_1)\right]. \tag{9}$$

The signs of $T - R$ and $S - P$ determine whether each strategy can invade the other when rare. Specifically, strategy 2 can invade strategy 1 when $T > R$, whereas strategy 1 can invade strategy 2 when $S > P$. These two invasion conditions partition the game space into four regimes (Figure 1B): prisoner's dilemma ($T > R$, $P > S$), hawk–dove game ($T > R$, $S > P$), stag–hunt game ($R > T$, $P > S$), and harmony game ($R > T$, $S > P$). In standard EGT, these outcomes are obtained by analyzing the stability of equilibrium conditions. For simple games, this analysis is straightforward.

A coexistence-theoretic approach recovers the same classification but provides a different interpretation (Figure 1C). For a two-strategy game determined by this payoff matrix, the niche difference and fitness difference are given by

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2}\right), \tag{10}$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \tag{11}$$

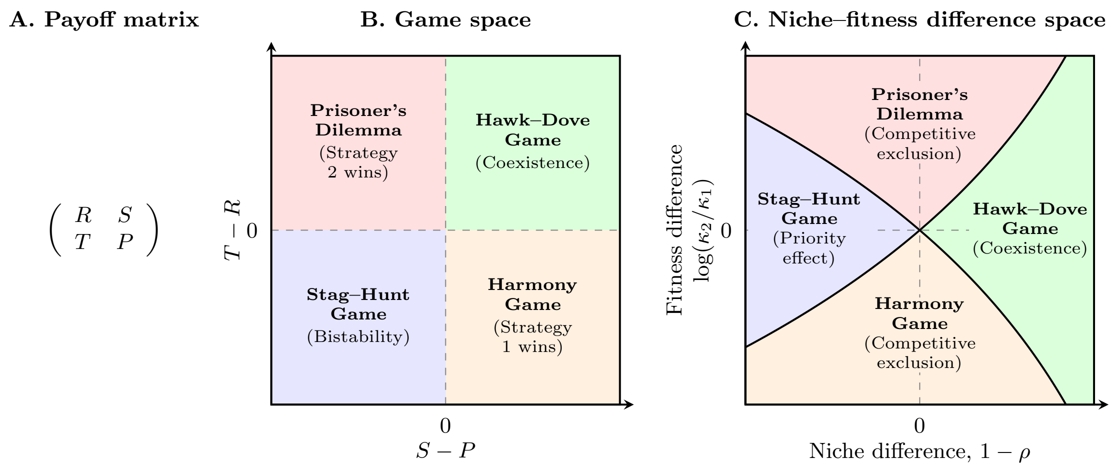

Figure 1: **Coexistence-theoretic interpretation of classic two-strategy games.** (A) General payoff matrix for a two-strategy game, where rows represent the focal strategy and columns represent the strategy of the interacting opponent. (B) Classification of four classic games based on the payoff differences $T - R$ and $S - P$. Prisoner's dilemma and harmony games lead to the dominance of one strategy; hawk–dove games allow coexistence; and stag–hunt games result in bistability. (C) Mapping of the same four games onto niche–fitness difference space. Hawk–dove games correspond to the coexistence regime, where stabilizing niche differences overcome fitness differences. Prisoner's dilemma and harmony games correspond to competitive-exclusion regimes, whereas stag–hunt games correspond to the priority-effect regime.

As shown above, mutual invasion requires $\rho < \kappa_2/\kappa_1 < 1/\rho$, which is equivalent to $T > R$ and $S > P$. This corresponds to the hawk–dove game, in which each strategy can invade when rare and the two strategies coexist. The remaining games also have natural interpretations in niche–fitness difference space. Prisoner's dilemma and harmony games represent competitive exclusion regimes, where one strategy has a fitness advantage that exceeds niche differences. In contrast, stag–hunt games represent a priority-effect regime, where neither strategy can invade the other and the outcome depends on initial conditions. Therefore, coexistence theory not only allows us to recover the classic results from two-strategy games but also provides new interpretations for these results in terms of niche and fitness differences. Building on this, we next apply this framework to microbial cooperation, where nonlinear benefits and environmental context can alter the coexistence of cooperators and defectors.

Microbial cooperation and public goods

Here, we consider a simple, phenomenological model of competition between cooperating and cheating yeast in a cooperative sucrose-metabolism system, where invertase production by cooperators allows cheaters to utilize glucose, a public good released through sucrose hydrolysis (Gore et al., 2009). Specifically, invertase production by cooperators incurs a cost c and generates a total benefit of 1 that is captured privately with efficiency ϵ. The payoffs for cooperators π_C and cheaters (defectors) π_D can then be written as

$$\pi_C(f) = \epsilon + f(1-\epsilon) - c, \tag{12}$$

$$\pi_D(f) = f(1-\epsilon), \tag{13}$$

where the use of glucose depends on the fraction of cooperators f. The dynamics of cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1-f)\left[\pi_C(f) - \pi_D(f)\right]. \tag{14}$$

For this model, cooperation is favored when efficiency is greater than the cost of cooperation, $\epsilon > c$, whereas defection is favored when the cost is greater than efficiency, $c > \epsilon$ (Gore et al., 2009).

Our framework allows us to reinterpret this result from the perspective of coexistence theory. In particular, the two strategies have no niche difference, corresponding to complete niche overlap, $\rho = 1$. In contrast, the fitness difference is determined by the balance between private efficiency and the cost of cooperation:

$$\frac{\kappa_D}{\kappa_C} = \exp(c - \epsilon). \tag{15}$$

Therefore, this model always predicts competitive exclusion, except in the neutral case $c = \epsilon$. The outcome is determined entirely by the fitness difference between cooperators and defectors, rather than by stabilizing niche differences.

When both payoffs include the nonlinear effect of glucose on microbial growth, cooperators and defectors can coexist:

$$\pi_C(f) = \left[\epsilon + f(1-\epsilon)\right]^\alpha - c, \tag{16}$$

$$\pi_D(f) = \left[f(1-\epsilon)\right]^\alpha, \tag{17}$$

where $\alpha = 0.15$ reflects the experimental relationship between glucose and microbial growth (Gore et al., 2009). From a coexistence-theoretic perspective, this means that nonlinear growth can generate sufficient niche difference to overcome the fitness difference. Indeed, we can show that the niche and fitness differences for this model are given by

$$1 - \rho = 1 - \exp\left(\frac{1 - \epsilon^\alpha - (1-\epsilon)^\alpha}{2}\right), \tag{18}$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(c + \frac{(1-\epsilon)^\alpha - 1 - \epsilon^\alpha}{2}\right). \tag{19}$$

Interestingly, the cost of cooperation does not affect niche difference. Instead, niche difference is generated by the concave relationship between glucose availability and microbial growth, which increases the relative benefit of the private glucose captured by cooperators when rare. When $\alpha = 1$, the niche difference completely disappears.

Supplementary Text

References

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